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Minimum Description Length

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Problem: Can we find a computable version of the induction principle?

Complexity theory had a big impact on Artificial Intelligence, especially in the domain of machine learning. Ray Solomonoff used it to solve *the problem of induction* in practice. His method, which merges Bayesianism and Kolmogorov complexity, mainly consists of complexity minimization. Though it led to strong theoretical results, it remains incomputable.

As an attempt to come closer to computability, one may restrict the set of 'models' that are considered to account for the data. This idea has been implemented in two principles: MDL (minimum description length, developed by Jorma Rissanen) and MML (minimum message length, developed by Chris Wallace).

In what follows, we present a version of MDL that we will call "crude MDL" or "ideal MDL" (which is very close to MML).

The Minimum Description Length principle



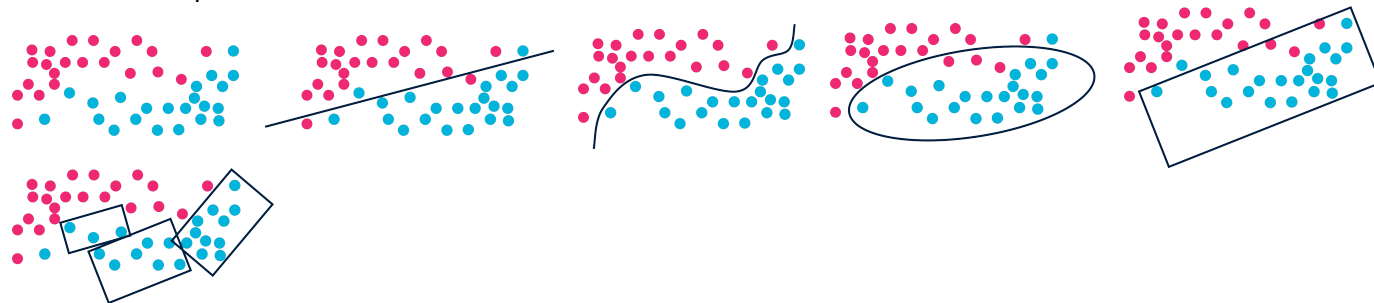
compressed version of the data if we accept to lose information.

- When it is taken from a set of possible causes, M_0 constitutes an explanation for the existence of the data D .

Conclusion:

The best model is neither the simplest one nor the one that offers the best fit with the data, but the model that achieves

Suppose you want to classify a set of N data $D = \{x_k\}$ in two classes, "blue"/"red". Your toolbox contains different computable models M_i that can be used to label the data as blue or red.



According to the (ideal) **Minimum Description Length principle**, you should choose the model that achieves the best compromise between simplicity and accuracy:

(ideal) MDL principle

Choose M_i such that

$$M_i = \operatorname{argmin}_j \{C(M_j) + \sum_k C(x_k|M_j)\}$$

Minimum description length 1

1/1 point (ungraded)

Suppose you don't know the true category (blue or red) of the N objects.
Among how many different classifications should we search the correct one?

○ 2 × N

☐ N^2

☒ 2^N



Answer

Correct:

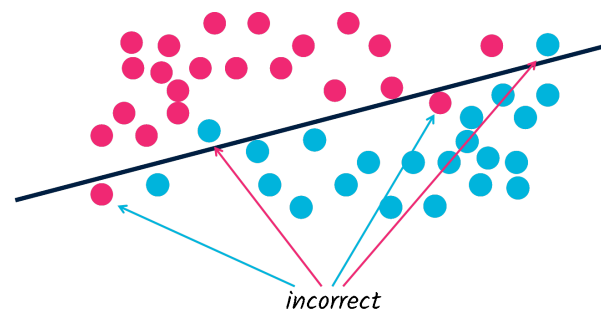
Any classification of data as "blue"/"red" corresponds to a function from D into $0, 1$. There are $2^{|D|}$ such functions (two possible labels for the first object, two for the second, and so on.)

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You have used 1 of 1 attempt

i Answers are displayed within the problem

Algorithmic Information Theory describes the learning task as a compression operation. The "**null model**" M_0 consists of remembering the class of each object. It requires N bits (1 bit for each element to be labelled as red or blue). We may take this as a baseline. The different models M_i are attempts to classify the data in a more economic way. In other words, the MDL problem consists of choosing among the $\{M_i\}$ the model that leads to **maximal overall compression**.



Maybe M_i is not perfect, which means that M_i classifies only n_i items among the $\{x_k\}$ correctly. We want to decide which is the "best" M_i .

Minimum description length 2

1/1 point (ungraded)

What is the compression (in number of bits saved by comparison with the null model) if M_i is perfect ($n_i = N$)?

☐ N

☐ $C(M_i)$

☒ $N - C(M_i)$

☐ $C(M_i) - N$



Answer

Correct:

If M_i is perfect, then the complexity of data is compressed from N (null model) down to $C(M_i)$. Compression amounts to: $N - C(M_i)$.

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i Answers are displayed within the problem

Suppose now that M_i isn't perfect ($n_i < N$). To reach a correct classification, one needs to designate all misclassified data, for instance by providing their indices.

Minimum description length 3

How much worse (in number of bits) is the compression of an imperfect model, as compared with the perfect case?

☐ $n_i \times \log_2 (N)$ bits

☐ $(N - n_i)$ bits

☐ $N \times \log_2 (n_i)$ bits

☒ $(N - n_i) \times \log_2 (N)$ bits

☐ $(N - n_i \times \log_2 (N))$ bits

☐ $N \times \log_2 (N - n_i)$ bits



Answer

Correct:

If M_i isn't perfect, then compression amounts to: $N - C(M_i) - C(I(i))$, where $I(i)$ is the list of indices of data that are wrongly classified by M_i . If these errors are randomly distributed, then $C(I(i)) \approx (N - n_i) \times \log_2 (N)$.

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You have used 1 of 1 attempt

i Answers are displayed within the problem

Note that the complexity of M_i includes the complexity of designating it among all models (this may amount to the complexity of i , i.e $\log_2(i)$ plus the complexity of all parameters values that may have to be determined within M_i).

Minimum description length 4

Can you provide a formal characterization of the *best* model in the M_i family?

☐ $\operatorname{argmin} (C(N) + C(M_i|N) + n_i \times \log_2 (N))$

☒ $\operatorname{argmin} (C(i) + C(M_i|i) + (N - n_i) \times \log_2 (N))$

☐ $\operatorname{argmin} (C(N) + C(M_i|i) + N \times \log((N - n_i)))$



Answer

Correct:

The best model number i is the solution of:
 $\operatorname{argmin} (C(M_i) + (N - n_i) \times \log_2 (N))$. It offers the best trade-off between its own complexity and the description of exceptions. We may also write:
 $\operatorname{argmin} (C(i) + C(M_i|i) + (N - n_i) \times \log_2 (N))$

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You have used 1 of 1 attempt

i Answers are displayed within the problem

Minimum description length 5

Can you describe how the worst model behaves (if we ignore its own complexity)?

- ☐ The worst model makes a random decision on each data.
- ☐ The worst model assigns the wrong class to each data.
- ☒ The worst model assigns the wrong class to half of the data.



Answer

Correct:

Worst models assign classes in a random way. They achieve negative compression! Note that a model that would systematically assigns the wrong class is an excellent model (if we ignore its own complexity), as the set of wrong data indices is $I(i) = [0, M]$, which as complexity 0 (knowing M).

The answer 'half of the data wrong' is correct as well (assuming the wrong data are randomly distributed).

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You have used 1 of 1 attempt

✓ Correct (1/1 point)

The MDL principle illustrates Gregory Chaitin's famous aphorism:

*“Comprehension is
compression”*

(Gregory Chaitin, 2004)

This means that the best theory is the one that achieves the best compression by reaching a compromise between its own complexity and the complexity of describing the facts that are left unexplained. This principle has far-reaching consequences in epistemology (read for instance *The symmetry and simplicity of the laws of physics and the Higgs boson*).

Suggested reading:

- Chaitin, G. J. (2004). On the intelligibility of the universe and the notions of simplicity, complexity and irreducibility. In W. Högerebe & J. Bromand (Eds.), *Grenzen und Grenzüberschreitungen, XIX*, 517-534. Berlin: Akademie Verlag.
- Maldacena, J. (2015). The symmetry and simplicity of the laws of physics and the Higgs boson. *European Journal of Physics*, 37 (1), 015802.
- Piantadosi, S. T. (2018). One parameter is always enough. *AIP Advances*, 8 (9).

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